1. A tech hub has 6 different project teams (P1 to P6) that need to meet tomorrow. However, there are only **3 meeting rooms** available (R1,R2,R3). Some teams have overlapping members and therefore cannot meet at the same time.

Constraints (Incompatibility List):

- P1 conflicts with: P2,P3,P6
- P2 conflicts with: P1,P3,P4
- P3 conflicts with: P1,P2,P5
- P4 conflicts with: P2,P6
- P5 conflicts with: P3,P6
- P6 conflicts with: P1,P4,P5

Task Requirements:

Initial State: Set the domain for every project variable to {R1,R2,R3}.

1. **Constraint Enforcement:** Apply AC-3 to this constraint graph.
2. **Step-by-Step Trace:** Provide a trace of the first 5 arc-reductions (e.g., "Arc (P1,P2) checked, no change").
3. **Consistency Check:** After running AC-3, determine if the problem is **arc-consistent**.

If we preemptively assign **Team 1** to **Room 1** , does AC-3 detect a failure, or does it leave a valid set of domains for the remaining rooms?

3. You are provided with a Sudoku puzzle classified as "Hard." Standard arc consistency often fails to solve such puzzles entirely but can significantly "thin out" the possibilities. Your task is to implement the AC-3 algorithm to see how much of the search space can be pruned before a human (or a backtracking algorithm) would need to start guessing.

Input Data: The following 81-character string representing the grid (0 denotes an empty cell):

```
0 0 0 0 0 6 0 0 0

0 5 9 0 0 0 0 0 8

2 0 0 0 0 8 0 0 0

0 4 5 0 0 0 0 0 0
```

```
0 0 3 0 0 0 0 0 0

0 0 6 0 0 3 0 5 0

0 0 0 0 0 7 0 0 0

0 0 0 0 0 0 0 0 0

0 0 0 0 5 0 0 0 2
```

Task Requirements:

1. **Arc Generation:** Automatically generate all 810+ binary constraints (arcs) based on Sudoku rules.
2. **State Tracking:** Output the number of values removed from all domains combined after the AC-3 algorithm terminates.
3. **Visualization:** Print the grid showing the remaining domain size for each cell (e.g., a "1" if solved, or "3" if three possibilities remain).

Key Question: Does AC-3 reduce any domain to zero (proving the puzzle unsolvable) or reduce all domains to one (solving it completely)?